KAIZEN MATHS

Lesson starter

Ratio and Proportion

Basic Ratios

Name _____

S1

Simplify the ratio 21:21 to its lowest terms.

S2

Divide £96 in the ratio 4:2. Find each share.

S3

Simplify the ratio 20:15 to its lowest terms.

S4

Divide £78 in the ratio 4:2. Find each share.

Show your working

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NUM-B9A2-L1 · S

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Exit ticket

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Simplify the ratio 26:28 to its lowest terms.

E2

Divide £180 in the ratio 5:4. Find each share.

E3

Simplify the ratio 14:18 to its lowest terms.

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Teacher answers & working

Ratio and Proportion · Basic Ratios

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S1

Simplify the ratio 21:21 to its lowest terms.

Answer:

1:1

Working:

Original ratio: 21:21

Find the highest common factor of 21 and 21.

$\mathrm{HCF}(21, 21) = 21$

Divide both parts by the same number so the ratio stays equivalent.

$21 \div 21 = 1$

$21 \div 21 = 1$

$21{:}21 = 1{:}1$

The simplified ratio keeps the same proportional relationship.

S2

Divide £96 in the ratio 4:2. Find each share.

Answer:

£64 and £32

Working:

Total ratio parts = 4 + 2 = 6.

Value of one part = £96 ÷ 6 = £16

First share = 4 × £16 = £64

Second share = 2 × £16 = £32

Shares are in the ratio 64:32, which simplifies to 4:2

Check: £64 + £32 = £96

Teacher answers & working

Ratio and Proportion · Basic Ratios

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S3

Simplify the ratio 20:15 to its lowest terms.

Answer:

4:3

Working:

Original ratio: 20:15

Find the highest common factor of 20 and 15.

$\mathrm{HCF}(20, 15) = 5$

Divide both parts by the same number so the ratio stays equivalent.

$20 \div 5 = 4$

$15 \div 5 = 3$

$20{:}15 = 4{:}3$

The simplified ratio keeps the same proportional relationship.

S4

Divide £78 in the ratio 4:2. Find each share.

Answer:

£52 and £26

Working:

Total ratio parts = 4 + 2 = 6.

Value of one part = £78 ÷ 6 = £13

First share = 4 × £13 = £52

Second share = 2 × £13 = £26

Shares are in the ratio 52:26, which simplifies to 4:2

Check: £52 + £26 = £78

Teacher answers & working

Ratio and Proportion · Basic Ratios

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E1

Simplify the ratio 26:28 to its lowest terms.

Answer:

13:14

Working:

Original ratio: 26:28

Find the highest common factor of 26 and 28.

$\mathrm{HCF}(26, 28) = 2$

Divide both parts by the same number so the ratio stays equivalent.

$26 \div 2 = 13$

$28 \div 2 = 14$

$26{:}28 = 13{:}14$

The simplified ratio keeps the same proportional relationship.

E2

Divide £180 in the ratio 5:4. Find each share.

Answer:

£100 and £80

Working:

Total ratio parts = 5 + 4 = 9.

Value of one part = £180 ÷ 9 = £20

First share = 5 × £20 = £100

Second share = 4 × £20 = £80

Shares are in the ratio 100:80, which simplifies to 5:4

Check: £100 + £80 = £180

Teacher answers & working

Ratio and Proportion · Basic Ratios

NUM-B9A2-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Simplify the ratio 14:18 to its lowest terms.

Answer:

7:9

Working:

Original ratio: 14:18

Find the highest common factor of 14 and 18.

$\mathrm{HCF}(14, 18) = 2$

Divide both parts by the same number so the ratio stays equivalent.

$14 \div 2 = 7$

$18 \div 2 = 9$

$14{:}18 = 7{:}9$

The simplified ratio keeps the same proportional relationship.